

ANTIMICROBIAL ACTIVITY OF SMALLER PROTEINS ISOLATED FROM WHITE SEEDS OF CAJANUS CAJAN

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ABSTRACT:

Two peptides from white seeds of *Cajanus cajan* were separated on Sephadex G-25 from acetone and ethanol prepared sample respectively. An antibacterial action of peptides were tested against *E. coli*, *K. aerogenes*, *Kl. pneumoniae*, *S. aureus* and *B. subtilis*. Peptide 1 and 2 in acetone and ethanol prepared sample of white seeds of *Cajanus cajan* showed inhibitory activity against Gram +ve and Gram -ve bacteria such as *E. coli*, *Kl. pneumoniae* and *B. subtilis*. However Peptide 1 and 2 in acetone and ethanol prepared sample had a notable inhibitory effect against fungi such as *Aspergillus niger*, *Aspergillus fumigatus*, and *Penicillium expansum*.

INTRODUCTION

In plant tissues, a series of dynamic defence mechanisms can be triggered upon wounding or perception of microorganisms. Newly formed carbohydrate can be deposited in the cell wall in response to penetration attempts by fungal hyphae (Aist, 1976) and preexisting cell wall proteins can be oxidatively cross linked upon wounding and elicitor treatment (Bradley et al., 1992). Both responses result in an induced fortification of the cell wall. Another strategy followed by plants to thwart invaders is based on the localized production of antimicrobial low molecular weight secondary metabolites known as phytoalexins (Van Elten et al., 1989, Maher et al., 1994). Moreover, the synthesis of many presumed defence related protein is induced when plants are confronted with pathogens (Linthrost, 1991).

A large number of shrubs, plants and seeds are extensively used by traditional healers (Hakeems) and herbal specialists to cure various diseases. A small protein, peptide and natural products (such as oils, steroids, alkaloids etc.) have been extracted which have been found to possess antimicrobial activity. Numerous studies have been carried out to extract natural product for screening antimicrobial activity (Rizki et al., 1987; Sheikh et al., 1985; Islam et al., 1989; Rasheed et al., 1990; Farzana & Khan, 1993) but there has been no report on small protein/ peptide isolation for biological activity from Pakistani shrubs, plants and seeds but during recent years

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greater attention has been focused to study small protein and peptide having antimicrobial activity have been isolated from plants and seeds (Robert & Selitrennikoff, 1990; Nair & Bruke, 1990; Robert et al., 1988; Robert & Selitrennikoff, 1986).

In this communication, we are reporting for the first time the isolation of small protein from white seeds of *Cajanus cajan* possessing antibacterial and antifungal activity. *Cajanus cajan* belongs to Fabaceae (Papilionaceae) family and commonly known as bean family. It is locally known as Manjhali and the seeds of *Cajanus cajan* are used as dal. It is widely distributed in Pakistan, India, Africa, east and west Indies.

MATERIALS AND METHODS

The seeds of *Cajanus cajan* were collected during September - November in dry state from the plants from Village Talib Noomrio, District Hyderabad, Pakistan.

Preparation of soluble extract:

10.0 grams of defatted residue of *Cajanus cajan* seeds were crushed in 33 ml cold distilled water and centrifuged at 4000 rpm. The supernatant was transferred to a 100 ml volumetric flask and this procedure was repeated twice and volume was made up to the mark with distilled water. Smaller proteins/peptides were precipitated by the addition of two fold ethanol or acetone. These isolated precipitates were dissolved in 0.2 M acetic acid and dialyzed overnight in the same.

Separation on Sephadex G-25:

The dialyzed ethanol or acetone precipitated fraction containing smaller protein/peptide was subject to a column of Sephadex G-25 (1.5 x 137 cm) and eluted with 0.2 M acetic acid. Fraction of 4.5 ml each was collected on fraction collector at a flow rate of 48 ml/hr.

Determination of Protein:

The absorbance at 280 nm was measured to monitor the protein during chromatography separation. The protein content of water extract was measured by the method of Lowry et al. (1951), using bovine serum albumin as a standard.

Electrophoresis:

The homogeneity of pooled fraction was checked by SDS disc gel electrophoresis by applying 50 µl of denatured sample on the surface of the gel. Power supply was adjusted at 5 ma current per tube for 80 minutes. The gel was stained with Coomassie blue R 250 and destained with methanol:acetic acid: water (30:60:10 V/V) as reported

by Davis (1964), Hames and Rickwood (1986).

Antibacterial activity:

The cultures of bacteria (*E. coli*, *Kl. aerogenes*, *Kl. pneumoniae*, *S. aureus*, and *B. subtilis*) grown overnight at 37°C were used for testing the antibacterial activity of different fractions separated on Sephadex G-25. The antibacterial activity was checked by seed plate method as reported by Rasheed et al. (1990). In this technique meat extract nutrient medium containing 1.5% agar was adjusted to pH 7.0, distributed in 40 ml quantity in screw capped bottles and sterilized. The bacterial culture was then added aseptically to the agar medium at 45°C, mixed well and poured immediately in sterilized petri-plates. After hardening, well were cut into agar and the *Cajanus cajan* seeds fractions were placed in these wells. The plates were incubated at 37°C and observations were made after 24 to 72 hours.

Antifungal activity:

Antifungal activity was tested against *Aspergillus niger*, *Aspergillus fumigatus*, *Aspergillus flavus* and *Penicillium expansum*. The diffusion plate method was used to test *Cajanus cajan* seeds fractions as reported by Reddish (1950) and Leven et al. (1979). In this technique, 0.1 ml of the fungal spore suspension (grown for 3 days in 10 ml of nutrient Dextrose agar) was thoroughly mixed with 20 ml of melted Sabouraud dextrose agar and poured into sterilized petri plates. When the agar was set, 5 holes of 6 mm diameter bore were made on each of the seeded plate. These holes were filled with 0.1 ml of the testing sample. All these experiments were performed in duplicate. The petri plates were incubated at 28°C for 7-8 days. All the culture plates were examined after 24-94 hours and the results are tabulated. The zone inhibition produced by the plant fractions were compared with zone produced by the standard.

RESULTS AND DISCUSSION

The results of protein content, anti-bacterial and antifungal activity from the water extract of white seeds of *Cajanus cajan* are summarize din Table 1. It is clearly noted that aqueous extract of *Cajanus cajan* seeds posses moderate antimicrobial activity against gram + ve, gram -ve and fungal species.

The initial steps of antibacterial/antifungal protein/peptide purification was done as described in materials and methods. Protein/peptide was isolated from the extract of ground *Cajanus cajan* seeds by precipitation with ethanol and acetone. The ethanol/acetone fractions were then separated by Sephadex G-25 column chromatography. Two optical density peaks were obtained as shown in Figure 1 and 2. The peptides associated with these optical density peaks from ethanol fraction were termed as EP1 and EP2 but peptide density pecks obtained from acetone fraction

were termed as AP1 and AP2 in the order, in which they were eluted from the column (Figure 1 & 2). The homogeneity of pooled fractions was checked by SDS polyacrylamide disc gel electrophoresis and were found homogeneous by showing single band (Figure 3).

Antibacterial/antifungal activities were assayed in EP1 and EP2 and both were found slightly active by producing clear zones of inhibition against a number of gram +ve and gram -ve bacterial such as *E. coli*, *Kl. pneumoniae* and *B. subtilis* and fungi including *Aspergillus niger*, *A. fumigatus* and *Penicillium expansum* as shown in Table 2. Peptide AP1 separated from acetone precipitated sample was found active against bacterias such as *E. coli* and *B. subtilis* and fungi including *A. niger*, *A. fumigaus* and *Penicillium expansum* but AP2 fraction showed strong inhibition effect against *Aspergillus nigher* and *Penicillium expansum*. This observation provide strong circumstantial evidence that anti-fungal peptide/protein play an important role in a plants antimicrobial defence system (Robert et al., 1988).

The result of present study confirm that the popular utilization of white sweeds of *Cajanus cajan* as an antimicrobial in skin and respiratory infection has some scientific justification. The extract of white seeds of *Cajanus cajan* can be used to prevent apple rot infection by *Penicillium expansum*. To our knowledge, this study is the first report on purification and characterization of antimicrobial peptide from white seeds of *Cajanus cajan* in this country. This study provides considerable scope in exploiting the local indigenous resources for isolation of antibacterial/antifungal peptides or smaller proteins. Further work is under progress in this Laboratory for isolation, purification and characterization of peptides/ smaller proteins for antibacterial/antifungal activity from other plants and results will be reported in near future.

ACKNOWLEDGMENT

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Table 1
Protein content and antimicrobial activity of crude
aqueous extract of *Cajanus cajan* white seeds

Protein content		1.05 mg/ml			
Antibacterial activity	E. coli	Kl. aerogenes	Kl. pneumoniae	S. aureus	B. subtilis
Inhibition zone	Trace	-ve	Trace	-ve	+ ve
Antifungal activity	Aspergillus niger	Aspergillus fumigatus	Aspergillus flavus	Penicillium expansum	
Inhibition zone	Trace	Trace	-ve	+ + ve	

Table 2
Antimicrobial activity of peptides isolated and purified from
white seeds of *Cajanus cajan* against bacteria and fungi

Peptides isolated with ethanol	Bacteria					Fungi			
	E. coli	Kl. aerogenes	Kl. pneumoniae	S. aureus	B. subtilis	Fumi.	Aspergillus niger	Aspergillus flavus	Penicillium exp.
P-1	-ve	-ve	Trace	-ve	-ve	-ve	Trace	-ve	+ve
P-2	Trace	-ve	-ve	-ve	Trace	Trace	Trace	-ve	+ve
Peptides isolated with acetone									
P-1	+ve	-ve	-ve	-ve	+ve	Trace	+ve	-ve	+++ve
P-2	-ve	-ve	-ve	-ve	Trace	Trace	+ve	-ve	+++ve

Negative = -ve; Positive =
 ++ = 10 mm
 +++ = 20 mm
 Trace = 2 mm; inhibitory zone;
 + = 5 mm inhibitory zone
 +++ = 15 mm inhibitory zone

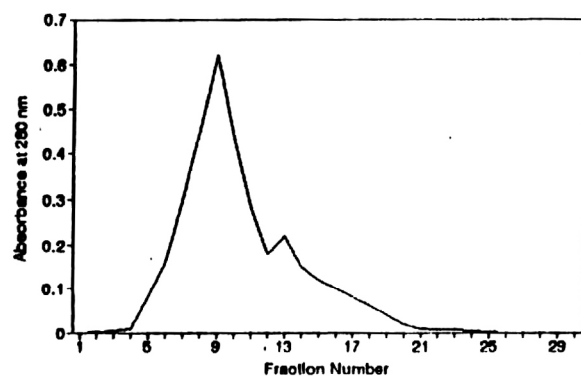


Figure 1: Separation of peptide on Sephadex G-25 from white seeds of *C. cajan* ppt. with ethanol.

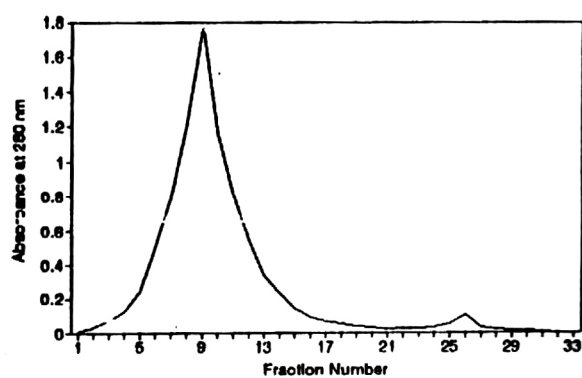


Figure 2: Separation of peptide on Sephadex G-25 from white seeds of *C. cajan* ppt. with acetone.



Figure 3: Electrophoresis pattern of crude and purified sample of *Cajanus cajan* seeds.

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